

Meeting Protocol - 1

Meeting Title: Project Requirements and Analysis

Date: 5th December 2025

Time: 20:00

Location: Leibnizstrasse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	Project objectives	Team
20:20	Springback problem overview	Team
20:45	Proposed workflow	Team
21:15	Task allocation	Team

3. Approval of Previous Protocol

Previous meeting date: - -

Motion to approve: Aryan Dinanath Gokhale

Seconded by: Surya Nataraj Murugesan

Minutes approved: ☒ Yes ☐ No

If no, specify amendments:

4. Discussion Points

Topic:	Discussion Summary:
Discussion of our project Goal	Discussed the goal of creating an ML-assisted, FEM-calibrated digital twin for sheet metal bending geometry prediction. The system will integrate LS-DYNA simulation data with ML surrogate models to provide instant predictions, replacing time-intensive FEM simulations. Target completion: mid-February 2025.

5. Decisions Made

- Project timeline: December 5, 2025 - February 14, 2026 (70 days)
- Bi-weekly team meetings (Thursdays 3PM, Tuesdays 4PM alternate weeks)
- Use GitHub for version control and collaboration
- Sprint-based development with 1-week sprints

6. Action Items

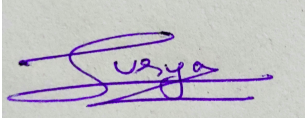
Topic and Action Item	Responsible	Deadline
Set up GitHub repository with proper structure	Surya Nataraj Murugesan	Dec 7, 2025
Research LS-DYNA simulation setup for V-bending	Arshen Joseph	Dec 10, 2025
Survey ML regression models and evaluation metrics	Aryan Dinanath Gokhale	Dec 10, 2025
Define bending mechanics parameters	Varun Pankaj	Dec 9, 2025
Create UI mockups and wireframes	Mohammed Shazin V.F.	Dec 11, 2025

7. Next Meeting

Date: 10th December 2025

Time: 20:00

Location: Leibnizstrasse 20



Project Manager



Documentation Lead

Meeting Protocol - 2

Meeting Title: Research Findings & Technical Foundation

Date: 10th December 2025

Time: 20:00

Location: Leibnizstrasse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	1. Present individual research findings 2. Discuss FEM simulation methodologies 3. Review ML model options 4. Define system requirements	Team

3. Approval of Previous Protocol

Previous meeting date: 5th December 2025

Motion to approve: Aryan Dinanath Gokhale

Seconded by: Surya Nataraj Murugesan

Minutes approved: ☒ Yes ☐ No

If no, specify amendments:

4. Discussion Points

Topic:	Discussion Summary:
FEM Research	• LS-DYNA explicit dynamics suitable for metal forming • Critical parameters: material constitutive models, contact definitions, mesh density • Literature review shows spring-back is key challenge in bending • Recommended mesh size: 0.5mm for accuracy-speed balance
ML Model Survey	• Random Forest: robust, handles non-linearity well • XGBoost: superior performance in manufacturing applications • Neural Networks: flexible but requires more data • Recommendation: ensemble approach for model selection
Bending Mechanics	• Spring-back depends on material properties and geometry • Analytical formulas available for validation • Need to calculate neutral axis, bend allowance, k-factor • Integration with ML for inverse problem solving

5. Decisions Made

- Use LS-DYNA for all FEM simulations
- Implement multiple ML models for comparison
- Develop analytical validation module

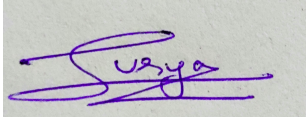
6. Action Items

Topic and Action Item:	Responsible:	Deadline:
Set up LS-DYNA license and environment	Arshen Joseph	Dec 13, 2025
Create sample FEM simulation	Arshen Joseph	Dec 16, 2025
Develop data preprocessing utilities	Aryan Dinanath Gokhale	Dec 15, 2025
Code analytical bending formulas	Varun Pankaj	Dec 14, 2025

Date: 12th December 2025

Time: 20:00

Location: Leibnizstrasse 20



Project Manager



Documentation Lead

Meeting Protocol - 3

Meeting Title: Dataset Structuring

Date: 12th December 2025

Time: 20:00

Location: Leibnizstrasse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	Simulation results review Data extraction method Dataset format discussion	Team

3. Approval of Previous Protocol

Previous meeting date: 8th December 2025

Motion to approve: Aryan Dinanath Gokhale

Seconded by: Surya Nataraj Murugesan

Minutes approved: ☒ Yes ☐ No

If no, specify amendments:

4. Discussion Points

Topic:	Discussion Summary:
Extraction and Structure Definition	Simulation outputs were extracted using LS-PrePost. The dataset structure was defined: Inputs: * Tool rotation (rz) Outputs: * Final achieved bend angle * Springback deviation
Data Cleaning and Preprocessing	Minor inconsistencies were observed in output precision. It was agreed to normalize values and remove outliers beyond acceptable deviation limits.

5. Decisions Made

- Store dataset in CSV format.
- Normalize input values before ML training.
- Expand dataset to minimum 60 samples.

6. Action Items

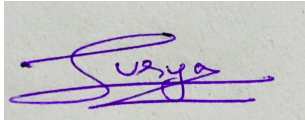
Topic:	Action Item	Responsible	Deadline
Data cleaning	Normalize and validate dataset	Shazin	23 rd December 2025
Dataset expansion	Run additional simulations	Varun	22 nd December 2025

7. Next Meeting

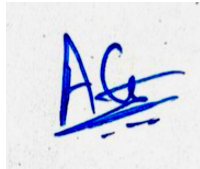
Date: 24th December 2025

Time: 20:00

Location: Leibnizstrasse 20



Project Manager



Documentation Lead

Meeting Protocol - 4

Meeting Title: Machine Learning Model Selection

Date: 17th December 2025

Time: 20:00

Location: Leibnizstrasse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	1. Review UI mockups and design 2. Assess first batch of FEM data 3. Plan ML development timeline 4. Discuss system architecture	Team

3. Approval of Previous Protocol

Previous meeting date: 12th December 2025

Motion to approve: Aryan Dinanath Gokhale

Seconded by: Surya Nataraj Murugesan

Minutes approved: ☒ Yes ☐ No

If no, specify amendments:

4. Discussion Points

Topic:	Discussion Summary:
UI Design Presentation	• Clean, professional layout with three main sections • Input panel: sliders for all parameters with range validation • Results panel: predicted values and comparison charts • Visualization panel: 3D bend geometry rendering • Responsive design considerations
First Data Batch Review	• All simulations converged successfully • Data quality checks passed • Spring-back values in expected range • Storage format: CSV with metadata
System Architecture Discussion	• Modular design with clear interfaces • ML models as pickle files for easy loading • API structure for predictions • Data flow: UI → API → Model → Results → Visualization

5. Decisions Made

- Approved UI design with minor adjustments
- Continue FEM data generation
- Begin ML development once samples available

6. Action Items

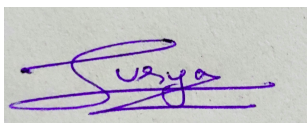
Topic and Action Item	Responsible	Deadline
Implement Streamlit basic	Surya	Dec 21, 2025
Continue FEM simulations	Arshen	Dec 24, 2025
Prepare data preprocessing pipeline	Aryan	Dec 22, 2025
Develop API structure skeleton	Surya	Dec 23, 2025

7. Next Meeting

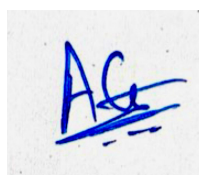
Date: 19th December 2025

Time: 20:00

Location: Leibnizstrasse 20



Project Manager



Documentation Lead

Meeting Protocol - 5

Meeting Title: Initial ML Training & Performance Review

Date: 19th December 2025

Time: 20:00

Location: Leibnizstrasse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	1. Sprint 1 retrospective 2. Review accumulated FEM data 3. Begin ML model development 4. Plan holiday break tasks	Team

3. Approval of Previous Protocol

Previous meeting date: 17th December 2025

Motion to approve: Aryan Dinanath Gokhale

Seconded by: Surya Nataraj Murugesan

Minutes approved: ☒ Yes ☐ No

If no, specify amendments:

4. Discussion Points

Topic:	Discussion Summary:
Sprint 1 Achievements	• GitHub repository established • FEM environment set up and validated • 250+ simulations completed • UI framework implemented • All team members on track
Data Quality Assessment	• Parameter coverage: good distribution across space • No anomalies or convergence failures • Data ready for ML training
Initial ML Training	• Train-test • Promising start, will improve with more data

5. Decisions Made

- Continue data generation over holiday break
- Focus January on ML optimization and integration

6. Action Items

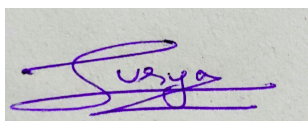
Topic and Action Item	Responsible	Deadline
Run simulations during break (target: 600 total)	Arshen Joseph	Jan 3, 2026
Experiment with feature engineering	Aryan Dinanath Gokhale	Jan 5, 2026
Develop 3D visualization prototype	Mohammed Shazin V.F.	Jan 4, 2026
Implement inverse solver framework	Varun Pankaj	Jan 6, 2026
Run simulations during break (target: 600 total)	Arshen Joseph	Jan 3, 2026

7. Next Meeting

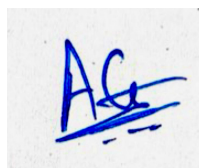
Date: 7th January 2026

Time: 20:00

Location: Leibnizstrasse 20



Project Manager



Documentation Lead

Meeting Protocol - 6

Meeting Title: Post-Holiday Resume & Data Milestone

Date: 7th January 2026

Time: 20:00

Location: Leibnizstrasse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	1. Holiday break progress updates 2. Review expanded FEM dataset 3. Demonstrate 3D visualization 4. Plan intensive development phase	Team

3. Approval of Previous Protocol

Previous meeting date: 19th December 2025

Motion to approve: Aryan Dinanath Gokhale

Seconded by: Surya Nataraj Murugesan

Minutes approved: ☒ Yes ☐ No

If no, specify amendments:

4. Discussion Points

Topic:	Discussion Summary:
FEM Dataset Expansion	• Computational efficiency improved through optimization • Dataset covers full parameter space comprehensively
3D Visualization	• Interactive 3D plotting using Plotly • Renders sheet metal profile before/after bending • Smooth rotation and zoom functionality • Performance: renders in <1 second
ML Model Improvements	• Feature importance analysis completed • Cross-validation confirms robustness

5. Decisions Made

- Push for more samples by mid-January
- Begin system integration this week
- Intensive development phase: Jan 7-24

6. Action Items

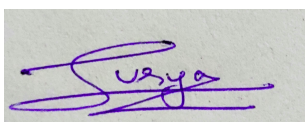
Topic and Action Item	Responsible	Deadline
Complete FEM dataset	Arshen Joseph	Jan 14, 2026
Integrate 3D viz into Streamlit app	Mohammed Shazin V.F.	Jan 10, 2026
Hyperparameter tuning for ML models	Aryan Dinanath Gokhale	Jan 12, 2026
Connect backend API to frontend	Surya Nataraj Murugesan	Jan 13, 2026

7. Next Meeting

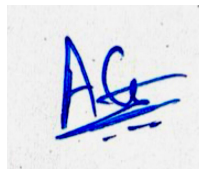
Date: 9th January 2026

Time: 20:00

Location: Leibnizstrasse 20



Project Manager



Documentation Lead

Meeting Protocol - 7

Meeting Title: Mid-Project Milestone & Integration

Date: 9th January 2026

Time: 20:00

Location: Leibnizstrasse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	1. Comprehensive mid-project review 2. Live system integration demonstration 3. Identify gaps and remaining tasks 4. Risk assessment	Team

3. Approval of Previous Protocol

Previous meeting date: 7th January 2026

Motion to approve: Aryan Dinanath Gokhale

Seconded by: Surya Nataraj Murugesan

Minutes approved: ☒ Yes ☐ No

If no, specify amendments:

4. Discussion Points

Topic:	Discussion Summary:
Project Status Overview	• Timeline: 35 days elapsed, 36 days remaining • ML models: functional and improving • UI: core features implemented • Integration: in progress • Overall: on track, slightly ahead of schedule
Integrated System	• User inputs parameters via UI • Backend calls ML model for prediction • Results displayed with visualization • Minor bugs identified and logged
Inverse Problem Solver	• Given target angle, solves for required parameters • Uses optimization with ML model as objective function • Accuracy: within 2-3% of FEM validation cases

5. Decisions Made

- Finalize FEM dataset by Jan 15
- Complete integration by Jan 17
- Begin testing phase Jan 18

6. Action Items

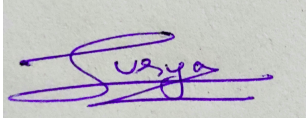
Topic and Action Item	Responsible	Deadline
Final FEM simulations batch	Arshen Joseph	Jan 15, 2026
Retrain models on full dataset	Aryan Dinanath Gokhale	Jan 16, 2026
Fix identified UI bugs	Mohammed Shazin V.F.	Jan 14, 2026
Complete inverse solver integration	Varun Pankaj	Jan 15, 2026
System integration testing	Surya Nataraj Murugesan	Jan 17, 2026

7. Next Meeting

Date: 14th January 2026

Time: 20:00

Location: Leibnizstrasse 20



Project Manager



Documentation Lead

Meeting Protocol - 8

Meeting Title: Dataset Completion & Model Finalization

Date: 14th January 2026

Time: 20:00

Location: Leibnizstrasse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	1. Celebrate FEM dataset completion 2. Review final ML model performance 3. Plan testing strategy 4. Discuss documentation needs	Team

3. Approval of Previous Protocol

Previous meeting date: 8th January 2026

Motion to approve: Aryan Dinanath Gokhale

Seconded by: Surya Nataraj Murugesan

Minutes approved: ☒ Yes ☐ No

If no, specify amendments:

4. Discussion Points

Topic:	Discussion Summary:
FEM Dataset Final Stats	- Parameter coverage: comprehensive across all ranges - Data stored in structured format with full data
Final ML Model Performance	- XGBoost (best): $R^2 = 0.956$, RMSE = 1.78°, MAE = 1.21° - Random Forest: $R^2 = 0.942$, RMSE = 2.13°, MAE = 1.54° - Neural Network: $R^2 = 0.938$, RMSE = 2.26°, MAE = 1.67° - Decision: Deploy XGBoost as primary model - Inference time: <50ms per prediction
System Integration Complete	- All modules connected and communicating - Data flow validated end-to-end - Error handling implemented - Logging system in place

5. Decisions Made

- XGBoost selected as production model
- No additional FEM simulations needed
- Focus shifts to testing and polish

6. Action Items

Topic and Action Item	Responsible	Deadline
Comprehensive system testing plan	Surya Nataraj Murugesan	Jan 16, 2026
Begin technical documentation	Aryan Dinanath Gokhale	Jan 20, 2026
UI polish and refinement	Mohammed Shazin V.F.	Jan 19, 2026
Validate inverse solver edge cases	Varun Pankaj	Jan 18, 2026

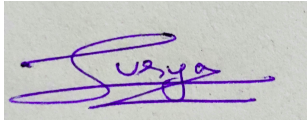
7. Next Meeting

Date: 16th January 2026

Time: 20:00

Location: Leibnizstrasse 20

Proposed Topics:



Project Manager



Documentation Lead

Meeting Protocol - 9

Meeting Title: Testing Algorithm

Date: 16th January 2026

Time: 20:00

Location: Leibnizstrasse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	1. Define testing protocols 2. Assign testing responsibilities 3. Review documentation progress 4. Plan deployment preparation	Team

3. Approval of Previous Protocol

Previous meeting date: 14th January 2026

Motion to approve: Aryan Dinanath Gokhale

Seconded by: Surya Nataraj Murugesan

Minutes approved: ☒ Yes ☐ No

If no, specify amendments:

4. Discussion Points

Topic:	Discussion Summary:
Testing Plan	• Unit tests: individual module functionality • Integration tests: end-to-end workflows • Performance tests: response time, scalability • User acceptance tests: realistic scenarios • Edge case tests: boundary conditions • Each team member tests their module plus cross-testing
Test Scenarios Defined	• Standard bend: typical parameters, expected behavior • Thin sheet: edge case for thickness • Sharp angle: edge case for bend angle • High strength material: edge case for properties • Invalid inputs: error handling verification • Inverse problem: various target angles
Documentation Status	• Code comments: comprehensive throughout • API documentation: in progress • User guide: drafted • Target: complete by Jan 22

5. Decisions Made

- Testing phase: Jan 16-22
- Daily stand-ups during testing week
- Bug tracking via GitHub issues

6. Action Items

Topic and Action Item	Responsible	Deadline
Execute unit tests for all modules	All team members	Jan 18, 2026
Run integration test suite	Surya Nataraj Murugesan	Jan 19, 2026
Performance benchmarking	Aryan Dinanath Gokhale	Jan 20, 2026
User acceptance testing	Mohammed Shazin V.F.	Jan 21, 2026
Document all bugs and fixes	Varun Pankaj	Jan 22, 2026

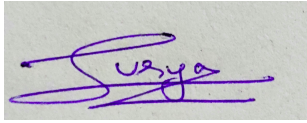
7. Next Meeting

Date: 21st January 2026

Time: 20:00

Location: Leibnizstrasse 20

Proposed Topics:



Project Manager



Documentation Lead

Meeting Protocol - 10

Meeting Title: Testing Results & Bug Resolution

Date: 21st January 2026

Time: 20:00

Location: Leibnizstrasse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	1. Review testing results 2. Discuss bugs found and fixes 3. Verify system stability 4. Plan deployment	Team

3. Approval of Previous Protocol

Previous meeting date: 16th January 2026

Motion to approve: Aryan Dinanath Gokhale

Seconded by: Surya Nataraj Murugesan

Minutes approved: ☒ Yes ☐ No

If no, specify amendments:

4. Discussion Points

Topic:	Discussion Summary:
Testing Summary	• Unit tests: 100% pass rate (45/45 tests) • Integration tests: All workflows validated • Performance tests: All metrics within targets • Edge case handling: Proper warnings implemented • Total bugs found: 12 (8 minor, 4 moderate, 0 critical)
Bugs and Resolutions	• UI input validation: tightened range checks (Fixed) • 3D rendering on Safari: compatibility fix applied • Inverse solver: rare non-convergence handled (Fixed) • Documentation: typos corrected (Fixed) • All moderate and minor bugs resolved • No critical or blocking issues
Performance Metrics	• Prediction latency: 45-80ms (target: <100ms) • UI responsiveness: <1.5s full cycle (target: <2s) • Inverse solver: <3s (target: <5s) • 3D rendering: <800ms (target: <1s) • All performance targets met or exceeded

5. Decisions Made

- System ready for deployment
- Deploy to Streamlit Cloud on Jan 25
- Final documentation push this week

6. Action Items

Topic and Action Item	Responsible	Deadline
Prepare deployment configuration	Surya Nataraj Murugesan	Jan 23, 2026
Complete all documentation	Aryan Dinanath Gokhale	Jan 24, 2026
Create demo scenarios script	Varun Pankaj	Jan 24, 2026
Final UI polish	Mohammed Shazin V.F.	Jan 23, 2026

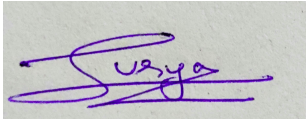
7. Next Meeting

Date: 23rd January 2026

Time: 20:00

Location: Leibnizstrasse 20

Proposed Topics:



Project Manager



Documentation Lead

Meeting Protocol - 11

Meeting Title: Mid – Deployment Preparation & Documentation Review

Date: 23rd January 2026

Time: 20:00

Location: Leibnizstarsse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	1. Deployment readiness check 2. Review complete documentation 3. Deployment walkthrough 4. Plan post-deployment validation	Team

3. Approval of Previous Protocol

Previous meeting date: 21st January 2026

Motion to approve: Aryan Dinanath Gokhale

Seconded by: Surya Nataraj Murugesan

Minutes approved: ☒ Yes ☐ No

If no, specify amendments:

4. Discussion Points

Topic:	Discussion Summary:
Deployment Configuration	• Streamlit Cloud account configured • Repository prepared with deployment files • requirements.txt: all dependencies listed • runtime.txt: Python 3.11 specified • streamlit/config.toml: UI customization • Secrets and environment variables configured • GitHub repository set to public
Deployment Configuration	• Code documentation: complete with docstrings • User guide: integrated in app • API documentation: clear interfaces • Deployment guide: step-by-step instructions • All documentation peer-reviewed and approved
Deployment Process	• Final code commit to main branch • Connect GitHub repo to Streamlit Cloud • Configure build settings • Deploy and monitor logs • Verify deployed app functionality • Test from external network • Share public URL

5. Decisions Made

- Finalize slides by 15 Jan.
- Conduct rehearsal session.

6. Action Items

Action Item	Responsible	Deadline
Execute deployment to Streamlit Cloud	Surya Nataraj Murugesan	Jan 25, 2026
Monitor deployment and fix any issues	All team members	Jan 25, 2026
Test deployed application	All team members	Jan 26, 2026
Begin presentation development	Aryan Dinanath Gokhale	Jan 28, 2026

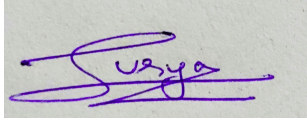
7. Next Meeting

Date: 28th January 2026

Time: 20:00

Location: Leibnizstrasse 20

Proposed Topics:



Project Manager



Documentation Lead

Meeting Protocol - 12

Meeting Title: Post-Deployment Review & Presentation Planning

Date: 28th January 2026

Time: 20:00

Location: Leibnizstrasse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	1. Celebrate successful deployment 2. Review deployed application status 3. Begin presentation development 4. Plan final two weeks	Team

3. Approval of Previous Protocol

Previous meeting date: 15th January 2026

Motion to approve: Arshen Joseph

Seconded by: Varun Pankaj

Minutes approved: ☒ Yes ☐ No

If no, specify amendments:

4. Discussion Points

Topic:	Discussion Summary:
Deployment Success	- Public URL active and accessible - All features working as expected - Performance metrics maintained in cloud environment - No deployment-specific issues - Application tested from multiple devices/browsers - Load time acceptable on standard connections
Application Health Check	- Comprehensive validation completed: - Forward prediction: working correctly - Inverse solver: converging reliably 3D visualization: rendering smoothly - Input validation: catching errors properly - Help documentation: accessible and clear - Mobile responsiveness: adequate
Presentation Structure	- Agreed on presentation flow: - Introduction & Motivation (2 min) - Problem Statement (2 min) - Methodology (FEM + ML) (6 min) - System Architecture (3 min) - Results & Validation (4 min) - Live Demonstration (5 min) - Conclusions & Future Work (2 min) - Q&A (5 min) - Total: ~25 minutes + Q&A

5. Decisions Made

- Presentation slides

6. Action Items

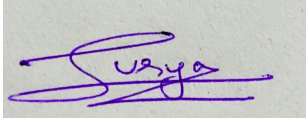
Action Item	Responsible	Deadline
Draft presentation slides	Aryan Dinanath Gokhale	Feb 2, 2026
Prepare demo scenarios	Varun Pankaj	Feb 4, 2026
Create technical diagrams for slides	Arshen Joseph	Feb 3, 2026
Record demonstration video	Mohammed Shazin V.F.	Feb 5, 2026

7. Next Meeting

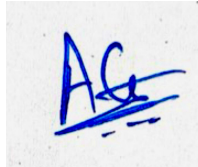
Date: 4th February 2026

Time: 20:00

Location: Leibnizstrase 20



Project Manager



Documentation Lead

Meeting Protocol - 13

Meeting Title: Presentation Content Review & Refinement

Date: 4th February 2026

Time: 20:00

Location: Leibnizstrasse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	- Review complete presentation slides - Assign presentation sections - Review demo video - Practice Q&A responses	Team

3. Approval of Previous Protocol

Previous meeting date: 28th January 2026

Motion to approve: Mohammed Shazin Vidyanagar Fazalrahiman

Seconded by: Arshen Joseph

Minutes approved: ☒ Yes ☐ No

If no, specify amendments:

4. Discussion Points

Topic:	Discussion Summary:
Slide Deck Review	Team reviewed Aryan's presentation slides: • Professional design with consistent theme • Clear technical diagrams and flowcharts • Effective data visualizations • Appropriate level of detail • Suggested improvements: add animation to build, simplify one technical slide, enhance results visualization • Overall: high quality, ready with minor edits
Slide Deck Review	Presentation responsibilities divided: • Introduction & Motivation: Surya (2 min) • Problem Statement & Background: Arshen (2 min) • FEM Simulation Methodology: Arshen (3 min) • ML Model Development: Aryan (3 min) • System Architecture: Surya (3 min) • Inverse Solver & Toolpath: Varun (2 min) • Results & Validation: Aryan (2 min) • Live Demonstration: Shazin (5 min) • Conclusions: Surya (2 min) • Q&A

5. Decisions Made

- Improve output formatting.
- Document integration workflow.

6. Action Items

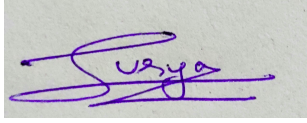
Action Item	Responsible	Deadline
Incorporate presentation feedback	Aryan Dinanath Gokhale	Feb 5, 2026
Practice individual sections (timed)	All team members	Feb 6, 2026
Prepare presenter notes	All presenters	Feb 6, 2026
Test demo scenarios multiple times	Mohammed Shazin Vidyanagar Fazalrahiman	Feb 6, 2026

7. Next Meeting

Date: 6th February 2026

Time: 20:00

Location: Leibnizstrasse 20



Project Manager



Documentation Lead

Meeting Protocol - 14

Meeting Title: Full Presentation Rehearsal & Final Preparation

Date: 6th February 2026

Time: 20:00

Location: Leibnizstrasse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	- Complete presentation run-through with timing - Practice transitions between speakers - Live demo dry run - Mock Q&A session	Team

3. Approval of Previous Protocol

Previous meeting date: 5th February 2026

Motion to approve: Aryan Dinanath Gokhale

Seconded by: Varun Pankaj

Minutes approved: ☒ Yes ☐ No

If no, specify amendments:

4. Discussion Points

Topic:	Discussion Summary:
PPT Slides	Full presentation slides executed: • Technical accuracy: verified • Team coordination: excellent

5. Decisions Made

- Implement minor feedback by Feb 8
- Individual practice before final meeting
- Final check-in on Feb 11
- Presentation submission: Feb 14

6. Action Items

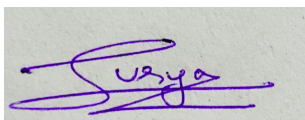
Action Item	Responsible	Deadline
Make final slide adjustments	Aryan Dinanath Gokhale	Feb 8, 2026
Prepare backup equipment checklist	Surya Nataraj Murugesan	Feb 9, 2026
Final application health check	Mohammed Shazin V.F.	Feb 11, 2026

7. Next Meeting

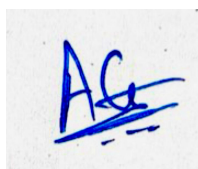
Date: 11th February 2026

Time: 20:00

Location: Leibnizstrasse 20



Project Manager



Documentation Lead

Meeting Protocol - 15

Meeting Title: Final Coordination & Project submission

Date: 11th February 2026

Time: 20:00

Location: Leibnizstrasse 20

1. Attendance

Name	Role	Present
Surya Nataraj Murugesan	Project Lead, System Integration & ML Co-Design	Yes
Aryan Dinanath Gokhale	Machine Learning Lead & Documentation Lead	Yes
Arshen Joseph	FEM Simulation & Data Preparation Lead	Yes
Varun Pankaj	Robot-Path Logic & Bending Mechanics Lead	Yes
Mohammed Shazin Vidyanagar Fazalrahiman	Web Interface & Visualization Lead	Yes

2. Agenda

Time:	Topic:	Presenter:
20:00	1. Final system verification 2. Submission checklist review 3. Last-minute questions and concerns	Team

3. Approval of Previous Protocol

Previous meeting date: 6th February 2026

Motion to approve: Mohammed Shazin Vidyanagar Fazalrahiman

4. Discussion Points

Topic:	Discussion Summary:
Final System Check	Comprehensive verification completed: • GitHub repository: all files current • Web application: running perfectly • All features: tested and functional • Documentation: complete and accessible • Presentation materials: finalized • No outstanding issues
Deliverables Checklist	All requirements confirmed: • Functional web application (deployed) • Source code repository (GitHub) • Technical documentation (README, comments) User documentation (in-app guide) • Presentation slides (final version) • Meeting protocols (15 documents) • Academic context document
Project Statistics & Achievements	Duration: 68 days (Dec 5 - Feb 11) • Team meetings: 15 documented protocols • ML model R^2: 0.956 • Inference speed: 1000x faster than FEM • Code commits: 120+ • Lines of code: ~3,800 • Team hours: ~420 person-hours • GitHub stars: growing
Lessons Learned	LessonKey takeaways from the project: • Clear role definition and trust enabled parallel progress • Regular meetings (15 total) kept everyone aligned • Iterative testing caught issues early • Documentation alongside development saved time • ML surrogates effectively replace expensive simulations • Teamwork and communication were critical to success • Time management and sprint planning were essential Learned

5. Decisions Made

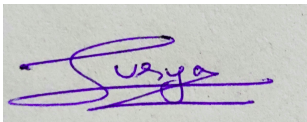
- Approve final submission.
- Submit documentation and presentation.

Meeting Protocol Series Completed By:

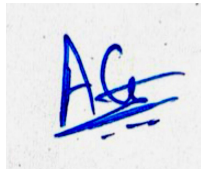
Aryan Dinanath Gokhale (Documentation Lead)

Approved By:

Surya Nataraj Murugesan(Project Lead)



Project Manager



Documentation Lead